MODULE 1
Explore generative AI

Module 1 Introduction: Maximize productivity with AI tools
Video • 1 min

Discover generative AI applications
Video • 2 min

Explore other ways AI can help in work
Video • 2 min

Understand how to write useful prompts
Reading • 10 min

AI models and the training process
Reading • 10 min

MODULE 2
Increase productivity with generative AI

MODULE 3
The importance of human involvement in AI

MODULE 4
Assistive AI tools to effectively use (Provided by Google Docs, Slides, Sheets, Meet and Gmail)

MODULE 5
Review: Maximize productivity with AI tools

AI models and the training process

Add files

Hey! we explored the way AI models power Artifacts. But how exactly are AI models made? And how can understanding this look and skills of a model help you unleash the full potential of AI tools?

Let's find out.

AI tools and AI models

Almost everything we use different things, even though they sound similar and are closely related. An **AI tool** is a powered software that uses one or more AI models with a user-friendly interface. An **AI model** is a complex program trained on vast amounts of data to recognize patterns and perform specific tasks. All AI tools are built on top of AI models, which are the brains for AI tools to work.

To better understand the relationship between an AI tool and an AI model, think about a car and its engine.



The car, with its user-friendly parts, like a steering wheel and big pedals, represents an AI tool, but it's helpless and useless when isolated from the engine, and car manufacturers don't sell engines to individuals. Similarly, AI models can't be used on their own. They're not designed for the end user to navigate. Additionally, they are wasteful to store. The same is true for an AI tool, with its interface and features, like the "engine" processing the information that you, the user, provide and showing the results to you.

But not all engines look the same. Similar to how some cars are better for certain things than others—for example, a truck is better for hauling heavy loads than a sports car—AI models are designed differently. Some are for generating text, images, videos, or even computer code. Having a basic idea of the specific function of the model you're using can be helpful as well.

If the car can also drive itself, navigating from one destination to another without you having to adjust the steering wheel, it would operate similar to how an AI agent operates. An **AI agent** is an AI-powered tool that can autonomously perform tasks on your behalf. For example, if you tell an AI agent to "Find me the best-rated, pet-friendly hotel in Los Angeles and book it," it can browse the web, read reviews, fill out forms, and complete the transaction. You set the goal, and the AI agent handles the steps.

Note: Some AI tools incorporate multiple AI models, working together to achieve more complex tasks. A tool might use one AI model to understand your request, another to generate the response, and a third to moderate the content. Think of it like a car: the engine provides the power, but it also relies on other components, like the transmission and brakes, to function properly.

The process of training AI models

AI engineers, who are responsible for the entire process of building AI models, play a crucial role in developing powerful and reliable AI tools. Let's look at how they train AI models, step by step.

1. **Define the problem to be solved.** First, AI engineers meet to understand exactly what they want the model to do. For example, they might decide they want to create a model that can predict customer churn (when customers stop using a service) or one that can identify different species of flowers in photos. This step is crucial because it helps them choose the right kind of data and the right type of model for the job.
2. **Collect relevant data to train the model.** AI engineers gather historical data (if they're trying to predict customer churn, they'll need data from past customers who have left and those who have stayed) or they create new data specifically for the task (if they're building a model to identify flowers, they'll need a large collection of labeled flower photos). The quality and quantity of this data are extremely important. The more diverse and representative the data, the better the model will be at handling different situations in the real world.
3. **Prepare the data for training.** AI engineers prepare the data by cleaning it (removing errors or inconsistencies), labeling it (adding tags to help the model understand what it's looking at), and splitting it into different sets. They typically use most of the data (around 70–80%) to train the model, a smaller portion (10–15%) to validate the model's performance during training, and another portion (10–15%) to test the model's performance after training is complete. This ensures that the model can generalize well to new, unseen data.
4. **Train the model.** AI engineers use specialized software and powerful computers to train the model. During training, the model learns by analyzing the training data and adjusting its internal parameters to minimize errors and improve its accuracy. This process can take anywhere from a few hours to several weeks, depending on the complexity of the model and the amount of data.
5. **Evaluate the model.** After training, AI engineers use the validation set (data that the model hasn't seen before) to assess how well the model performs on new, unseen data. They look at metrics like accuracy (how often the model makes correct predictions), precision (how many of the positive predictions were actually correct), and recall (how many of the actual positive cases the model identified). They may also test the model on a separate "test set" for a final, unbiased evaluation.
6. **Refine and retrain the model.** Based on the evaluation results, AI engineers may go back and adjust the model's parameters, add more data, or even change the model's architecture (the overall structure of the model). They repeat the training and evaluation process until they're satisfied with the model's performance.
7. **Deploy the model.** Once the AI engineers are satisfied with their model's performance, they deploy it, or in other words, they make it available for use in real-world applications. This might involve integrating the model into a software application, a website, or a cloud platform.

Model training is an iterative process. AI engineers rarely get it right on the first try. They often need to go through multiple cycles of refining the data, adjusting the model, and retraining it to achieve the desired results.

Even after deployment, monitoring the model's performance is crucial. Over time, the real-world data that the model encounters might change, a phenomenon known as "data drift." This can cause the model's accuracy to decline. AI engineers need to continuously monitor the model's performance and retrain it with new data as needed to maintain its effectiveness.

Finally, it's important to remember that AI models can sometimes reflect biases present in the data they're trained on. If the training data doesn't accurately represent the real world or if it contains historical biases, the model may perpetuate those biases in its predictions. AI engineers are increasingly aware of this issue and are working to develop techniques to mitigate bias and ensure that AI models are fair and equitable. This is an ongoing area of research and development.

Now, you understand how AI models are made, and you can choose the right AI tool for your needs. Let's see how you can use AI tools to increase your productivity in your daily work.

Mark as completed

Like Dislike Report an issue

Go to next item